

Web Application Firewall: 3 Types of WAF and Key Capabilities

A Web Application Firewall (WAF) is a security system that monitors and filters incoming traffic to a web application. It is designed to protect web applications from attacks by blocking malicious traffic and allowing legitimate traffic to pass through.

WAFs are typically deployed in front of a web server and use a set of rules to analyze incoming traffic to identify and block potentially malicious requests. They can be used to protect against a variety of threats, such as cross-site scripting (XSS), SQL injection, and cookie poisoning. WAFs can be configured to block traffic based on a variety of criteria, such as the IP address of the requesting client, the type of request being made, and the payload of the request.

This is part of a series of articles about [application security](#).

Why Is WAF Security Important?

Web Application Firewall (WAF) security is important because it can help to protect web applications from a variety of threats. Web applications are often targeted by hackers because they contain sensitive information, such as customer data, financial records, and intellectual property. If a web application is compromised, it can result in data breaches, financial losses, and damage to a company's reputation.

WAF security is particularly important because web applications are vulnerable to a wide range of attacks, including:

- **XSS attacks:** Inject malicious code into a web application, which is then executed by the victim's web browser. WAFs can protect against XSS attacks by blocking traffic that contains malicious code or by normalizing requests and responses to remove the code.
- **SQL injection:** Inject malicious code into a web application's database through a SQL query. WAFs can protect against SQL injection attacks by blocking traffic that contains malicious code or by normalizing requests and responses to remove the code.

- **Cookie poisoning:** Manipulates the cookies that are used to store information about a user's session with a web application. WAFs can protect against cookie poisoning attacks by blocking traffic that contains malicious code or by normalizing requests and responses to remove the code.
- **Denial of Service (DoS) attacks:** Overwhelm a web application with traffic in an attempt to make it unavailable to legitimate users. WAFs can protect against DoS attacks by limiting the amount of traffic that is allowed to reach the web application and by blocking traffic that is identified as being part of a DoS attack.
- **Malware:** WAFs can protect against malware by blocking traffic that is known to be associated with malware or by scanning incoming traffic for signs of malware.
- **Brute force attacks:** Attempt to guess a user's password by trying a large number of possible combinations. WAFs can protect against brute force attacks by blocking traffic that is identified as part of a brute force attack.

These types of attacks can allow hackers to gain unauthorized access to a web application, steal sensitive data, or manipulate the application to perform unintended actions.

By implementing a WAF, organizations can protect their web applications from these types of threats and reduce the risk of a data breach or other security incident. WAFs can also help to improve the overall security posture of an organization by providing an additional layer of defense against cyber threats.

What Is the Difference between WAF and Firewall?

A WAF and a firewall are both types of security systems that are designed to protect networks and systems from external threats. However, there are some key differences between the two:

Scope of protection

A firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules. It is designed to protect the entire network, including servers, devices, and other systems. A WAF, on the other hand, is specifically designed to protect web applications from threats. It monitors and filters incoming traffic to a web application and is typically deployed in front of a web server.

Type of threats

A firewall is primarily designed to protect against network-based threats, such as malware, viruses, and denial of service (DoS) attacks. A WAF is specifically designed to protect against

web application-based threats, such as cross-site scripting (XSS), SQL injection, and cookie poisoning.

Deployment

Firewalls can be deployed at various points in a network, such as at the perimeter of a network, between a network and the Internet, or within a network. WAFs are typically deployed in front of a web server.

Configuration

Firewalls are typically configured using rules that specify which types of traffic are allowed or denied based on factors such as the source and destination IP addresses and port numbers. WAFs are typically configured using a set of rules that specify which types of traffic should be allowed or blocked based on a variety of criteria, such as the IP address of the requesting client, the type of request being made, and the payload of the request.

Types of Web Application Firewalls Solutions

Network-Based WAFs

Network-based WAFs are deployed at the perimeter of a network and are designed to protect all web applications on the network. They operate by inspecting incoming traffic to the network and blocking any traffic that does not meet the configured security rules. Network-based WAFs are typically deployed on hardware devices or as a software solution that runs on a dedicated server.

Pros:

- Can protect all web applications on a network
- Can be used to protect against a wide range of threats, including network-based threats
- Can be configured to block traffic based on a variety of criteria, such as IP addresses and port numbers

Cons:

- Requires a dedicated hardware or software solution
- May require a significant investment in hardware and maintenance
- May not provide as granular a level of control as host-based WAFs

Host-Based WAFs

Host-based WAFs are deployed on individual web servers and are designed to protect the web application running on that server. They operate by inspecting incoming traffic to the

web application and blocking any traffic that does not meet the configured security rules. Host-based WAFs are typically deployed as software solutions that run on the web server.

Pros:

- Provides a granular level of control over the web application being protected
- Can be deployed on any type of web server
- Does not require a dedicated hardware solution

Cons:

- Only protects the web application running on the server where it is deployed
- May require additional resources to manage and maintain

Cloud-Hosted WAFs

Cloud-hosted WAFs are WAFs that are hosted and managed by a third-party provider. They operate by inspecting incoming traffic to a web application and blocking any traffic that does not meet the configured security rules. Cloud-hosted WAFs are typically deployed as a service, with the WAF provider managing the hardware and software infrastructure required to run the WAF.

Pros:

- No need to purchase or maintain hardware or software infrastructure
- Easy to scale up or down as needed
- Can be used to protect web applications hosted on any type of server

Cons:

- Requires a subscription to a third-party service
- May not provide as much control over the WAF configuration as on-premises solutions
- May not provide the same level of protection as an on-premises WAF, depending on the provider and the specific service being used.

What Are the Key Capabilities of WAF Tools?

Some of the main capabilities and features of WAFs include:

Traffic filtering

WAFs are designed to inspect incoming traffic to a web application and block any traffic that

does not meet the configured security rules. This can help to protect against a variety of threats, such as XSS and SQL injection.

Request and response normalization

WAFs can normalize incoming requests and outgoing responses to ensure that they conform to a defined set of rules. This can help to prevent attacks that rely on manipulating the structure of requests or responses.

Encryption and decryption

WAFs can encrypt and decrypt incoming and outgoing traffic to protect sensitive data in transit. This can help to prevent attackers from intercepting and stealing sensitive data.

Signature-based detection

WAFs can use a database of known attack signatures to identify and block malicious traffic. This can help to protect against known threats and vulnerabilities.

Anomaly detection

WAFs can use machine learning algorithms to identify and block traffic that deviates from normal patterns. This can help to protect against zero-day threats and other unknown threats.

Integration with other security systems

WAFs can be integrated with other security systems, such as intrusion detection and prevention systems (IDPS) and security information and event management (SIEM) systems, to provide a more comprehensive security solution.

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